

A short handbook for people who want to brew better coffee at home. It covers what happens during extraction, how to set the ratio of coffee to water, how grind size and water temperature change the cup, and how to taste your results.

1 Extraction

1.1 What Extraction Means

Brewing coffee is extraction: hot water dissolves soluble compounds from ground coffee. Roughly 30% of the mass of a roasted bean can dissolve in water, but a pleasant cup extracts only about 18 to 22% of the dry coffee. Acids dissolve first, then sugars, and the bitter, woody compounds come last.

When too little is extracted (under-extraction), the cup tastes sour, thin and salty. When too much is extracted (over-extraction), it tastes bitter, hollow and drying. The goal is the balanced middle, where the sweetness of the sugars covers the acidity and the bitterness stays in the background.

1.2 Strength Is Not Extraction

Strength describes how concentrated the drink is: how much dissolved coffee ends up in the cup. It is measured as total dissolved solids (TDS). Filter coffee usually tastes best between 1.15% and 1.35% TDS. Strength and extraction are separate: a strong cup can still be under-extracted, and a weak cup can be over-extracted. You change strength mostly with the ratio of coffee to water, and extraction mostly with grind size, temperature and contact time.

2 Ratio and dose

2.1 The Brew Ratio

The brew ratio is the weight of coffee compared to the weight of water. A ratio of 1:16 means one gram of coffee for every sixteen grams of water. Most filter recipes sit between 1:15 and 1:17; 1:16 is a good starting point. A lower number (1:15) gives a stronger cup, a higher number (1:17) a lighter one.

Weigh both coffee and water with a kitchen scale. Measuring by volume with spoons is unreliable, because a spoonful of a light roast weighs more than a spoonful of a dark roast.

2.2 Calculating A Dose

To find the water for a dose, multiply the coffee weight by the ratio. For 20 g of coffee at 1:16 you need 320 g of water (20×16). To brew a 500 g pot at 1:16, divide instead: $500 / 16$ gives about 31 g of coffee. One gram of water is one millilitre, so the scale reading in grams equals the volume in millilitres.

3 Grind size

3.1 Why Grind Size Matters

Grinding exposes the inside of the bean to water. A finer grind has more surface area, so water extracts faster; a coarser grind extracts more slowly. Grind size also controls how fast water flows through a filter: fine grounds slow the flow and lengthen the contact time.

As a rule of thumb, adjust grind size first when a cup tastes off. If it is sour, grind finer. If it is bitter, grind coarser. Change one thing at a time so you can taste the effect.

3.2 Matching Grind To Method

Each brewing method suits a range of grind sizes. Espresso needs a very fine grind, like flour. A pour-over cone such as a V60 works with a medium-fine grind, similar to table salt. A French press, where coffee steeps for about four minutes, needs a coarse grind, like sea salt, so that the metal filter can hold the grounds back. A burr grinder gives more even particles than a blade grinder, and even particles extract evenly.

4 Water

4.1 Temperature

Water between 90 and 96 degC extracts coffee well. Water just off the boil, around 96 degC, suits light roasts, which are dense and harder to extract. Dark roasts extract easily and can taste harsh with very hot water, so 90 to 92 degC works better for them. Without a thermometer, let boiled water rest for about 30 seconds before pouring.

4.2 Water Quality

A cup of filter coffee is about 98% water, so the water itself matters. Use clean, odour-free water. Very soft water can make coffee taste flat, while very hard water dulls acidity and leaves scale in kettles. Filtered tap water is a good choice in most places.

5 Brewing a pour-over

5.1 The Bloom

Start by pouring about twice the weight of the coffee in water, for example 40 g of water on 20 g of coffee, and wait 30 to 45 seconds. Freshly roasted coffee releases carbon dioxide when it gets wet, and the bed of coffee bubbles and rises. This is called the bloom. Letting the gas escape first helps the rest of the water extract the coffee evenly.

5.2 Pouring And Timing

After the bloom, pour the rest of the water slowly in circles, keeping the bed evenly wet. For a single cup with 20 g of coffee and 320 g of water, the whole brew should take about three minutes. If it runs much faster, grind finer next time; if it runs much slower, grind coarser.

6 Tasting

6.1 Tasting Your Cup

Let the coffee cool a little before tasting, because flavours are easier to notice when the cup is warm rather than hot. Notice three things: acidity (bright, sour or flat), sweetness and bitterness, and body (how heavy the coffee feels in your mouth). Write down the recipe and what you taste so you can repeat a good cup.

6.2 Adjusting The Next Brew

Use what you taste to change one variable. Sour and thin points to under-extraction: grind finer or use hotter water. Bitter and drying points to over-extraction: grind coarser or use cooler water. If the cup is balanced but too weak or too strong, change the ratio rather than the grind.